

Network design practice 6

김경민

June 9th, 2026

1 Assignment 1

```
10 - 8000:8000
11 depends_on:
12   - influxdb
13 environment:
14   - INFLUXDB_DATABASE=influx b3
15   - INFLUXDB_HOST=influxdb
16   - INFLUXDB_PASSWORD=admin
17   - INFLUXDB_PORT=8086
18   - INFLUXDB_USERNAME=admin
19   - NS3_HOST=192.168.48.132
20
21 influxdb:
22   image: influxdb:1.8-alpine
```

Changed ip address to the virtual machine ip 192.168.48.132.

```

[NEAR-RIC]: Loading SM ID = 143 with def = RLC_STATS_V0
[NEAR-RIC]: Loading SM ID = 146 with def = TC_STATS_V0
[NEAR-RIC]: Loading SM ID = 145 with def = SLICE_STATS_V0
[NEAR-RIC]: Loading SM ID = 3 with def = ORAN-E2SM-RC
[NEAR-RIC]: Loading SM ID = 142 with def = MAC_STATS_V0
[NEAR-RIC]: Loading SM ID = 2 with def = ORAN-E2SM-KPM
[iApp]: Initializing ...
[iApp]: nearRT-RIC IP Address = 127.0.0.1, PORT = 36422
[NEAR-RIC]: Initializing Task Manager with 2 threads
[E2AP]: E2 SETUP-REQUEST rx from PLMN 131.313 Node ID 838860800 RAN type ngran_gNB
[NEAR-RIC]: Accepting RAN function ID 3 with def = ORAN-E2SM-RC
[NEAR-RIC]: Accepting RAN function ID 2 with def = ORAN-E2SM-KPM
[E2AP]: E2 SETUP-REQUEST rx from PLMN 131.313 Node ID 822083584 RAN type ngran_gNB
[NEAR-RIC]: Accepting RAN function ID 3 with def = ORAN-E2SM-RC
[NEAR-RIC]: Accepting RAN function ID 2 with def = ORAN-E2SM-KPM
[E2AP]: E2 SETUP-REQUEST rx from PLMN 131.313 Node ID 855638016 RAN type ngran_gNB
[NEAR-RIC]: Accepting RAN function ID 3 with def = ORAN-E2SM-RC
[NEAR-RIC]: Accepting RAN function ID 2 with def = ORAN-E2SM-KPM
[E2AP]: E2 SETUP-REQUEST rx from PLMN 131.313 Node ID 872415232 RAN type ngran_gNB
[NEAR-RIC]: Accepting RAN function ID 3 with def = ORAN-E2SM-RC
[NEAR-RIC]: Accepting RAN function ID 2 with def = ORAN-E2SM-KPM
[E2AP]: E2 SETUP-REQUEST rx from PLMN 131.313 Node ID 889192448 RAN type ngran_gNB
[NEAR-RIC]: Accepting RAN function ID 3 with def = ORAN-E2SM-RC
[NEAR-RIC]: Accepting RAN function ID 2 with def = ORAN-E2SM-KPM

Position of UE with IMSI 14 is 1259.75:2578.4:0 at time 0.2, UE connected to Cell: 4
Position of UE with IMSI 15 is 2663.98:2185.57:0 at time 0.2, UE connected to Cell: 3
Total energy consumption for mmWave cell 2: 291.318J at time 0.3, diff from last measurement is: 118.754J
Total energy consumption for mmWave cell 3: 579.306J at time 0.3, diff from last measurement is: 268.545J
Total energy consumption for mmWave cell 4: 512.263J at time 0.3, diff from last measurement is: 226.973J
Total energy consumption for mmWave cell 5: 353.918J at time 0.3, diff from last measurement is: 148.21J
Position of UE with IMSI 1 is 2901.06:1777.98:0 at time 0.3, UE connected to Cell: 3
Position of UE with IMSI 2 is 2447.85:1591.17:0 at time 0.3, UE connected to Cell: 3
Position of UE with IMSI 3 is 1761.97:2397.77:0 at time 0.3, UE connected to Cell: 4
Position of UE with IMSI 4 is 1905.84:1060.27:0 at time 0.3, UE connected to Cell: 5
Position of UE with IMSI 5 is 1425.76:2746.21:0 at time 0.3, UE connected to Cell: 4
Position of UE with IMSI 6 is 2244.52:1917.03:0 at time 0.3, UE connected to Cell: 2
Position of UE with IMSI 7 is 1412.01:2111.15:0 at time 0.3, UE connected to Cell: 4
Position of UE with IMSI 8 is 2571.11:2363.37:0 at time 0.3, UE connected to Cell: 3
Position of UE with IMSI 9 is 2507.32:1347.54:0 at time 0.3, UE connected to Cell: 1
Position of UE with IMSI 10 is 1239.87:1990.59:0 at time 0.3, UE connected to Cell: 1
Position of UE with IMSI 11 is 1620.25:1635.3:0 at time 0.3, UE connected to Cell: 5
Position of UE with IMSI 12 is 2819.99:2344.28:0 at time 0.3, UE connected to Cell: 3
Position of UE with IMSI 13 is 2581.91:1801.92:0 at time 0.3, UE connected to Cell: 3
Position of UE with IMSI 14 is 1259.55:2578.66:0 at time 0.3, UE connected to Cell: 4
Position of UE with IMSI 15 is 2663.93:2185.2:0 at time 0.3, UE connected to Cell: 3

```

Figure 1: nearRT RIC terminal, python3 gui.trigger.py terminal

Successfully run RIC-TaaP GUI.



Figure 2: Monitoring on firefox localhost

The monitor shows that the maximum energy consumption is 375,416J, and the current energy consumption is also 375,416J. This indicates that the maximum energy consumption capacity is saturated and utilized to the full.

2 Assignment 2

```
=== Evaluating Cells for Energy Saving ===  
Cell 2 meeting the Action Defination Condition  
  
=== Checking meeting condition and start reporting ===  
  
=== Evaluating Cells for Energy Saving ===  
Cell 2 meeting the Action Defination Condition  
Serving Cell 5 - UE 4: -4.00 dB  
Updated neighbor cell 3 for UE 4: SINR -16.00 (sample 2/5)  
Updated neighbor cell 4 for UE 4: SINR -19.00 (sample 2/5)  
Serving Cell 5 - UE 11: 15.00 dB  
Updated neighbor cell 4 for UE 11: SINR -11.00 (sample 2/5)  
Updated neighbor cell 3 for UE 11: SINR -8.67 (sample 3/5)  
  
=== Checking meeting condition and start reporting ===  
  
=== Evaluating Cells for Energy Saving ===  
Cell 2 meeting the Action Defination Condition
```

Figure 3: xApp handling power usage of the cell

xApp locating the cell and reporting the power consumption. It updates the neighboring cells accordingly.

```

=== Target Cell Selection for UE 6 ===
Current Cell: 2
Evaluating Cell 3: SINR 6.00 dB
Found better cell: 3 (SINR: 6.00 dB)
Evaluating Cell 5: SINR -6.00 dB
Selected Target Cell: 3 (SINR: 6.00 dB)
[xApp]: data.toTargetCell= 3 ..
[xApp]: Send Handover Control message to move IMSI 6 from cellId 2 to target cellId 3
[xApp]: CONTROL-REQUEST tx
[xApp]: CONTROL ACK rx
[xApp]: Successfully received CONTROL-ACK
[xApp]: Handover request sent successfully to node 0
[xApp]: CONTROL-REQUEST tx
[xApp]: CONTROL ACK rx
[xApp]: Successfully received CONTROL-ACK
[xApp]: Handover request sent successfully to node 1
[xApp]: CONTROL-REQUEST tx
[xApp]: CONTROL ACK rx
[xApp]: Successfully received CONTROL-ACK
[xApp]: Handover request sent successfully to node 2
[xApp]: CONTROL-REQUEST tx
[xApp]: Successfully received CONTROL-ACK
[xApp]: Handover request sent successfully to node 2
[xApp]: CONTROL-REQUEST tx
[xApp]: CONTROL ACK rx
[xApp]: Successfully received CONTROL-ACK
[xApp]: Handover request sent successfully to node 3
[xApp]: CONTROL-REQUEST tx
[xApp]: CONTROL ACK rx
[xApp]: Successfully received CONTROL-ACK
[xApp]: Handover request sent successfully to node 4
All handovers complete (1/1) or timeout reached for cell 2
[xApp]: Send switch off Control message to switch off cell 2
[xApp]: CONTROL-REQUEST tx
[xApp]: CONTROL ACK rx
[xApp]: Successfully received CONTROL-ACK

```

Figure 4: xApp handling power usage of the cell

Now, with the xApp is running, in the ns3 running terminal it is finding a cell with a better SINR(6dB) and initiates handover.



Figure 5: monitor with xApp on

The monitor indicates that the maximum energy is 468,411J, and the current consumption is 373,952J. It runs beneath the maximum capacity due to power saving.